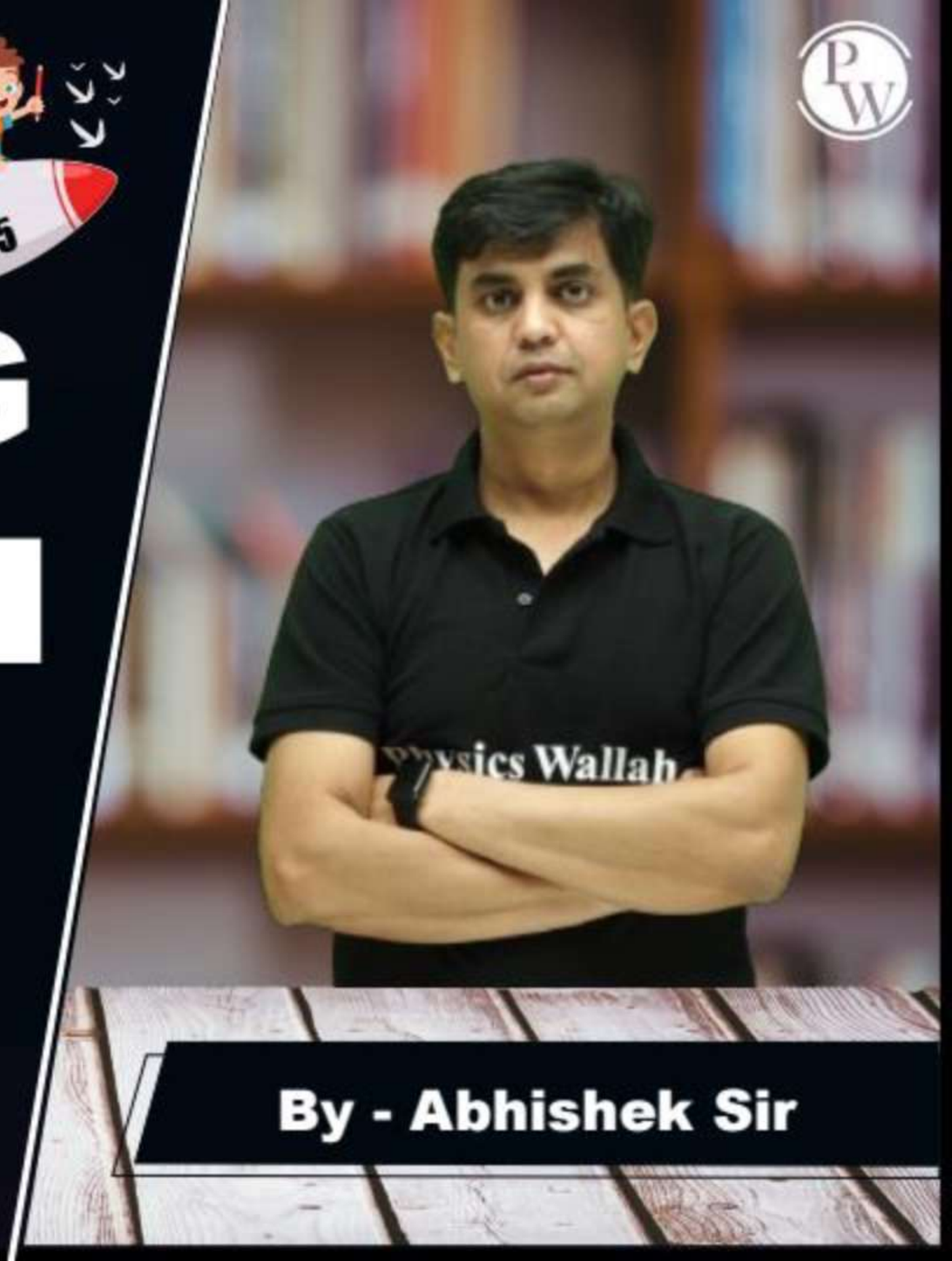
CS & IT ENGINEERING

Computer Network

Application Layer

Lecture No. - 01



By - Abhishek Sir

Recap of Previous Lecture

Topic TCP Timers

Topics to be Covered

Topic Application Layer Protocol

Topic DNS

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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#Q. Assume that the bandwidth for a TCP connection is 1048560 bits/sec. Let α be the value of RTT in milliseconds (rounded off to the nearest integer) after which the TCP window scale option is needed. Let β be the maximum possible window size with window scale option. Then the values of α and β are.

[GATE-2015, Set-2, 2-Mark]

(A) 63 milliseconds 65535×2^{14}

(B) 63 milliseconds 65535×2^{16}

(C) 500 milliseconds 65535×2^{14}

(D) 500 milliseconds 65535×2^{16}

Ans: C

Bandwidth = 1048560 bits/sec

$$\alpha = \frac{(2^{16}-1)\text{ bytes}}{\text{Bandwidth}} = \frac{(2^{16}-1)*8\text{ bits}}{1048560\text{ bits/sec}} = 0.5\text{ sec}$$

$$= 0.5*10^3\text{ ms} = 500\text{ ms}$$

$$B = (2^{16}-1)*2^{14}\text{ bytes} = 65535*2^{14}\text{ bytes}$$

#Q. Suppose two hosts use a TCP connection to transfer a large file. Which of the following statements is/are False with respect to the TCP connection?

I. If the sequence number of a segment is m, then the sequence number of the subsequent segment is always m+1. FALSE

II. If the estimated round trip time at any given point of time is t sec, the value of the retransmission timeout is always set to greater than or equal to t sec. TRUE

III. The size of the advertised window never changes during the course of the TCP connection. FALSE

IV. The number of unacknowledged bytes at the sender is always less than or equal to the advertised window TRUE

[GATE-2015, Set-1, 1-Mark]

(A) III only

(B) I and III only

(C) I and IV only

(D) II and IV only

Ans: B

[MSQ]

#Q. A TCP server application is programmed to listen on port number P on host S. A TCP client is connected to the TCP server over the network. Consider that while the TCP connection was active, the server machine S crashed and rebooted. Assume that the client does not use the TCP keepalive timer. Which of the following behaviors is/are possible?

[GATE-2021, Set-1, 2-Mark]

(A) If the client was waiting to receive a packet, it may wait indefinitely TRUE

(B) The TCP server application on S can listen on P after reboot TRUE

(C) If the client sends a packet after the server reboot, it will receive a RST segment TRUE

(D) If the client sends a packet after the server reboot, it will receive a FIN segment FALSE

Ans: A, B & C

TCP
Client
1) Socket
2) Connect

Port
P

TCP
Server

Host
S

1) Socket
2) Bind
3) Listen
4) Accept

TCP Client

Server Crash (reboot)

Port P	TCP Server	Host S

1) Socket
2) Bind
3) Listen

Topic : Application Layer

-> Network application program (process)

[e.g. Browser]

–> Port Number (16-bit) : Identifier

Topic : Two Process Communication

-> Two network processes, belongs to different hosts

-> Communicate over network, by exchanging "messages"

Application layer PDU = "Message"

Topic : Protocol

–> Set of Rules

–> Define, how data is exchanged

Topic : Application Layer

Application Layer Protocols :

- ② DNS : Domain Name System
- ① HTTP : Hyper Text Transfer Protocol
- ④ FTP : File Transfer Protocol
- ③ SMTP : Simple Mail Transfer Protocol

Email

POP / IMAP

Topic : Two Process Communication

Two paradigm :

1. **Client - Server**
   [e.g. Web browsing, Email]
2. **Peer to peer**
   [e.g. BitTorrent, VoIP]

Topic : Client - Server Paradigm

Server :

→ Always on host

→ Permanent IP Address

→ Mostly in data centers

Client :

→ Communicate (contact) with server

→ May have dynamic IP address

→ Clients do not communicate directly with each other

Client : Web browser

Server : Web Server

Topic : Peer - Peer Architecture

→ No always on server

→ Arbitrary end systems directly communicate

→ Complex management (Peers can change IP address)

→ e.g P2P file sharing (BitTorrent), Voice over IP (VoIP)

Topic : DNS

→ Domain Name System (DNS)

→ 'Stateless' Application Layer Protocol

→ Used to map between 'IP Address' and 'Host Name' (vice versa)

Topic : DNS

[DNS server]

→ Distributed Database : Implemented in hierarchy of many name servers

→ Hosts (DNS clients) and DNS server communicate to resolve names [Address / name translation]

→ Used to translate 'Host Name' into 'IP Address' [vice versa]

Topic : DNS Hierarchy

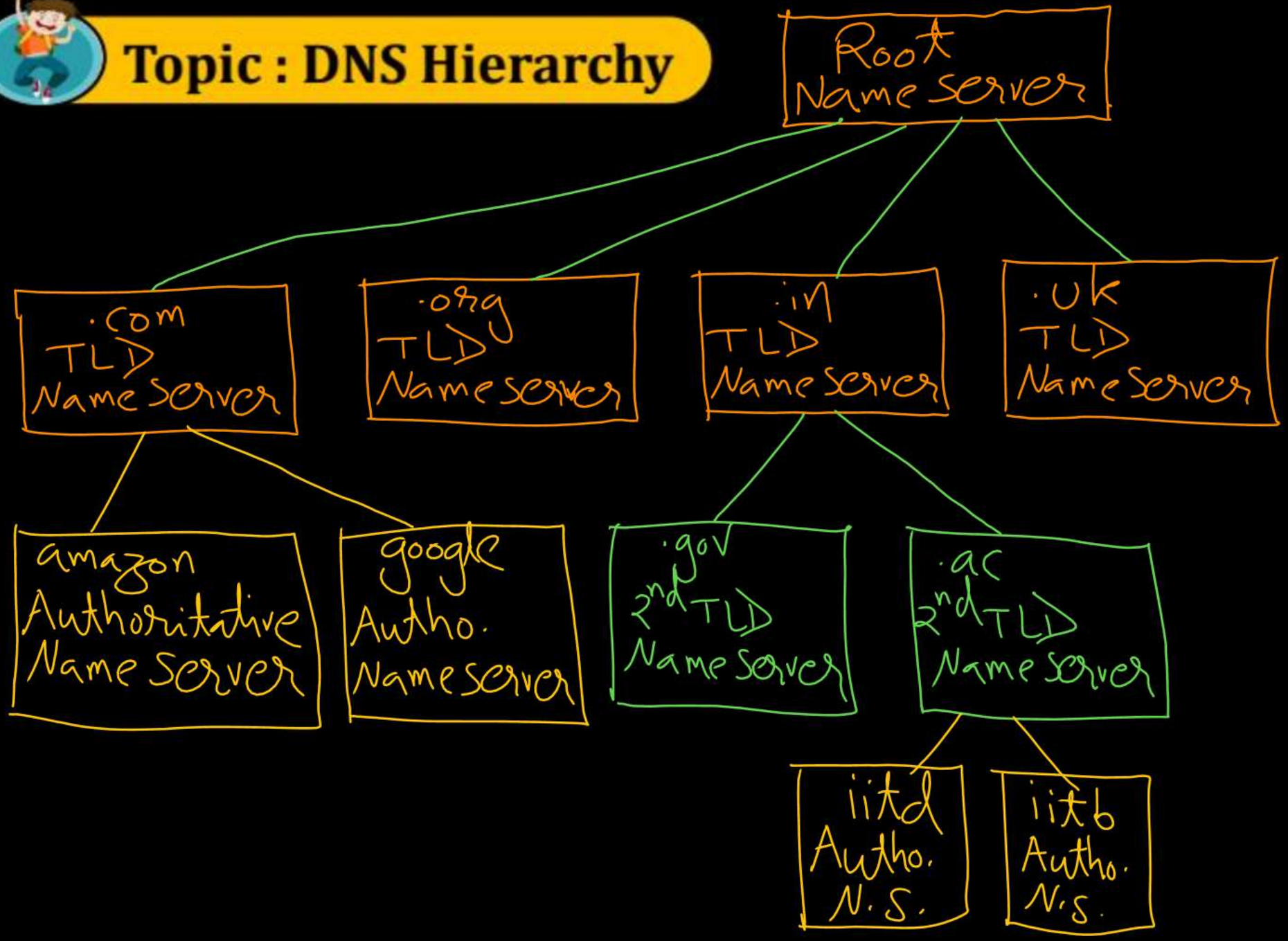
→ Distributed Hierarchical Database

→ Global Hierarchy

→ DNS name servers :

1. Root Name Servers [Point of Contact]
2. Top-Level Domain (TLD) Name Servers
3. Authoritative Name Servers

Topic : DNS Hierarchy



Topic : Root Name Servers

→ ICANN manages root DNS domain [Root Name Server]

→ 13 logical root name servers world wide
[each server replicated many times]

Topic : TLD Name Servers

→ Responsible for all top level domains like ".com", ".org" etc
[Country domains e.g. ".in", ".uk", etc]

Topic : Authoritative Name Servers

→ Organization's own DNS Server(s)

→ Providing authoritative host name to IP mappings

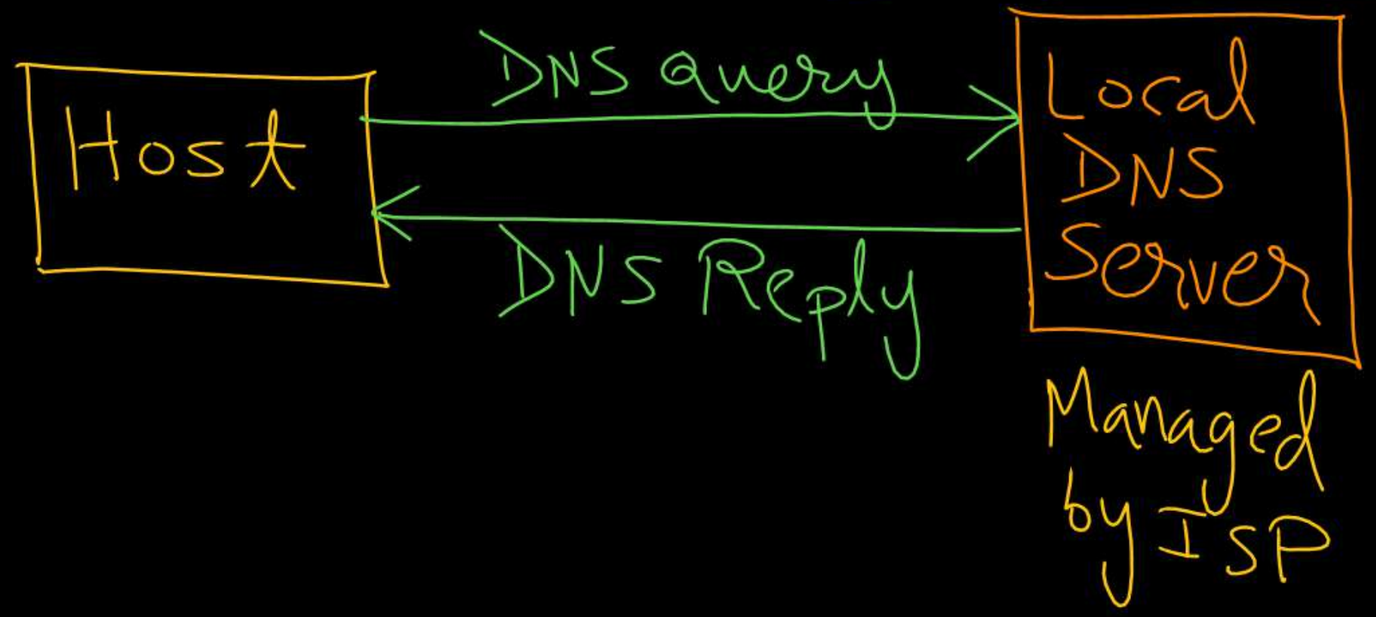
→ Can be maintained by organization or service provider

Topic : Local DNS Name Servers

→ When host make DNS query, it is sent to its local DNS server

→ Local DNS server returns reply

→ if Local DNS server does not have valid name-address pair then it forward request into DNS hierarchy for resolution



Topic : Local DNS Name Servers

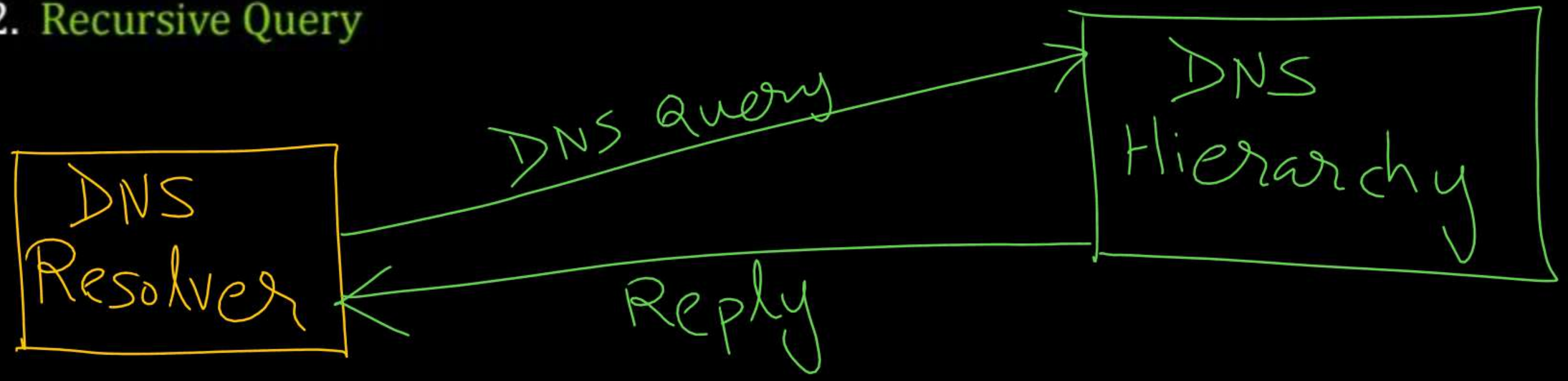
→ Each ISP has local DNS name server

→ Local DNS server does not belongs to DNS hierarchy

Topic : DNS Name Resolution

→ DNS Name Resolution or Lookup :

1. Iterated Query
2. Recursive Query



2 mins Summary

Topic Application Layer Protocol

Topic DNS

THANK - YOU